

RCE Cluster Execution Guide

gRPC Assignment

This guide provides a basic template for running the gRPC assignment on the RCE cluster. Students are expected to run the **gRPC server on one compute node** and one or more **clients on other compute nodes**.

1. Request Compute Nodes

Login to the RCE cluster and request an interactive allocation:

```
salloc --nodes=3 --ntasks-per-node=1
```

After the allocation is granted, check the nodes assigned to you:

```
scontrol show hostnames $SLURM_JOB_NODELIST
```

For example, if the allocated nodes are node01, node02, and node03, use node01 as the server and node02/node03 as clients.

2. Start the gRPC Server

Open a terminal connected to the server compute node:

```
ssh node01
cd ~/HW3/problem1
python3 server.py
```

The server should listen on a port accessible from other compute nodes, for example 0.0.0.0:50051. Keep the server running while clients are using the system.

3. Start Client 1

Open another terminal and connect to the second compute node:

```
ssh node02
cd ~/HW3/problem1
python3 client.py node01:50051
```

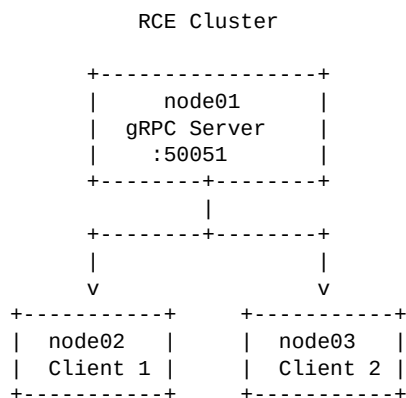
The client should connect to the gRPC server and provide the required command-line interface. Client operations trigger gRPC requests to the server.

4. Start Client 2

Open a third terminal and connect to another allocated compute node:

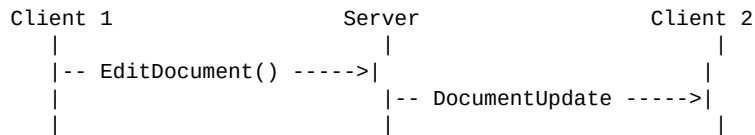
```
ssh node03
cd ~/HW3/problem1
python3 client.py node01:50051
```

The resulting setup is:



5. Demonstrating Problem 1

Students should demonstrate multiple clients interacting with the same document server. For example, Client 1 creates a document and Client 2 subscribes to updates. Client 1 then performs an edit while Client 2 is subscribed.



6. Running Problem 2

Use the same approach for the Food Ordering System. For example: node01 = server, node02 = customer client, and node03 = restaurant client.

```

ssh node01
cd ~/HW3/problem2
python3 server.py

ssh node02
cd ~/HW3/problem2
python3 customer.py node01:50051

ssh node03
cd ~/HW3/problem2
python3 restaurant.py node01:50051 "Pizza House"
  
```

Students should demonstrate the order progressing through:

```
PLACED -> ACCEPTED -> PREPARING -> READY
```

7. Important Notes

Use Compute Nodes: Run the server and clients on compute nodes allocated through Slurm, not on the login node.

Use the Correct Server Address: Clients must connect to the hostname and port where the server is running, such as node01:50051. Do not use localhost:50051 when client and server are on different nodes.

Keep the Server Running: The server process must remain active while clients are making requests.

Multiple Clients: For the demonstration, students should run at least two client processes simultaneously and show that they communicate with the same server.

Port Number: Port 50051 is an example. Students may use another available port if required by the cluster configuration.

8. Quick Reference

```

# Allocate nodes
salloc --nodes=3 --ntasks-per-node=1

# Find allocated nodes
scontrol show hostnames $SLURM_JOB_NODELIST

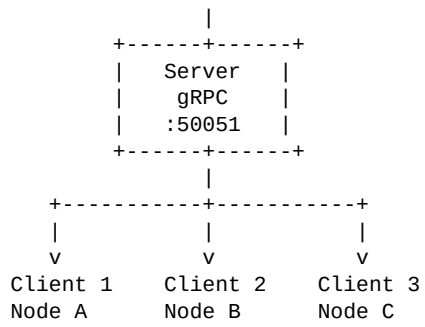
# Server
ssh <server-node>
cd <project-directory>
python3 server.py

# Client 1
ssh <client-node-1>
cd <project-directory>
python3 client.py <server-node>:50051

# Client 2
ssh <client-node-2>
cd <project-directory>
python3 client.py <server-node>:50051
  
```

9. Expected Architecture

Compute Cluster



The important requirement is that the server and clients are separate processes and communicate using gRPC over the cluster network.